

TECHNICAL ROADMAP TEMPLATE



Submission Deadline: May 17, 2026

Format: PDF (Strictly maximum 5 pages)

Naming Convention: [Track]_[University]_[TeamName]_Roadmap.pdf

TEAM INFORMATION

Team Name	USJR - FlowerBoys ASEAN
Institution	University of San Jose-Recoletos
Country	Philippines
Track	[✓] Smart Cities
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SECTION 1: EXECUTIVE SUMMARY (PROBLEM-SOLUTION FIT)

Public transit in ASEAN cities — particularly Cebu, Philippines — operates with minimal real-time visibility on either side of the vehicle door. Commuters cannot determine a vehicle's occupancy before it arrives, nor predict arrival times. Drivers routinely exceed legal passenger capacity while operators lack demand data for efficient fleet allocation. Traffic congestion costs the Philippine economy PHP 3.5 billion daily in lost productivity, a crisis worsened by absent live monitoring that enables illegal overloading (*sabit*) and weakens safety compliance enforcement (JICA, 2018; LTFRB, 2024). In Cebu, commuters already report peak-period wait times exceeding 20 minutes, and existing transit technology consistently bypasses low-income passengers who depend on jeepneys most.

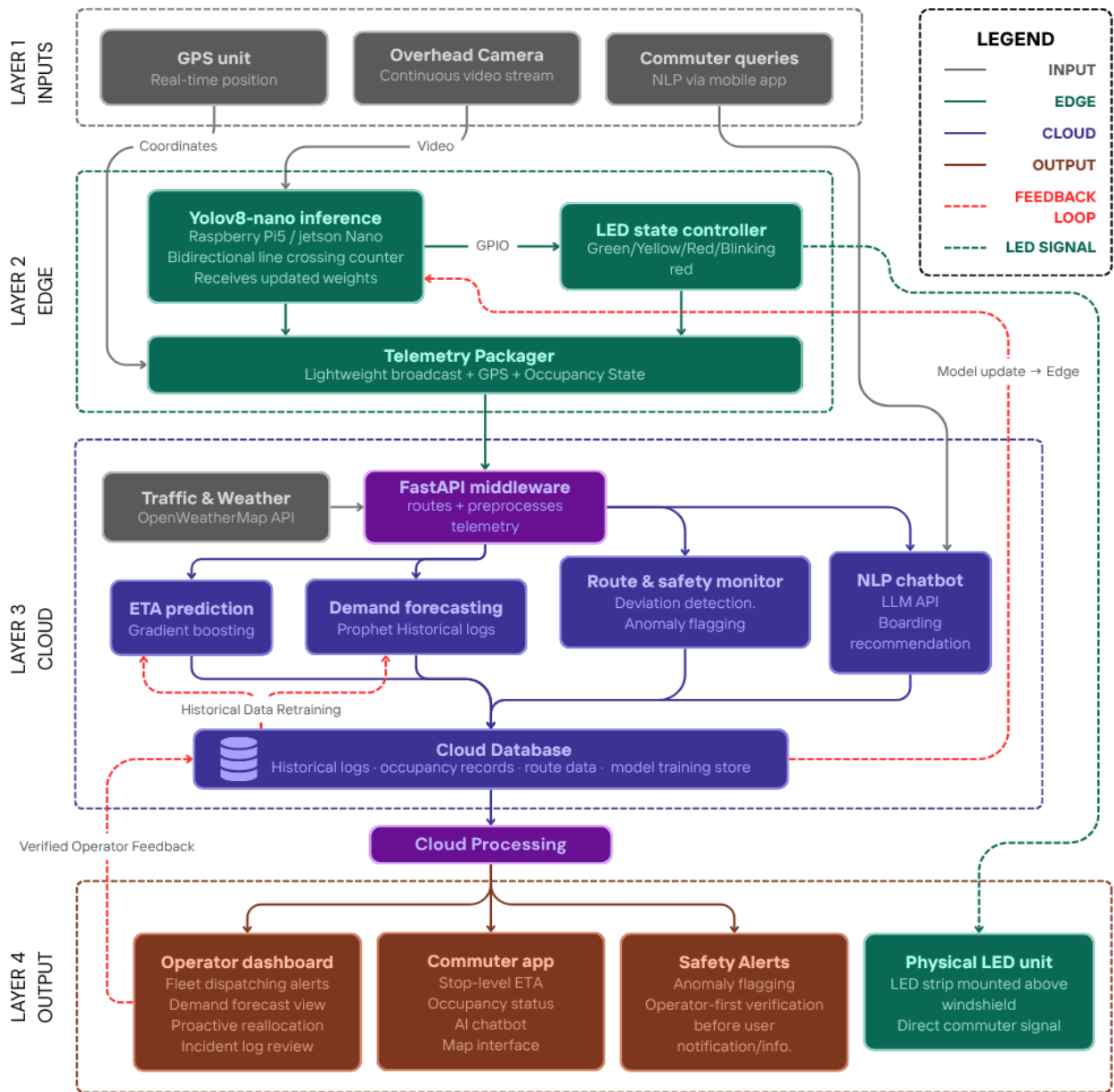
LoadSense is a dual-layer intelligent transportation platform designed to close this information gap. The first layer is an in-vehicle edge AI system: an overhead camera runs YOLOv8-nano offline, performing bidirectional passenger counting and classifying occupancy into four tiers — Green (available), Yellow (filling), Red (at capacity), and Blinking Red (overloaded) — displayed via a windshield LED strip visible to waiting commuters. The second layer is a cloud intelligence system: GPS telemetry feeds a server that predicts arrival times, detects route deviations and driving anomalies, and forecasts demand. Designed to retrofit into traditional jeepneys and modern PUVs without fleet replacement, LoadSense targets SDG 9 and SDG 11.

SECTION 2: TECHNICAL ARCHITECTURE

2.1 System Components:

Inputs	Processing Core	Outputs
- Overhead camera video stream (in-vehicle, continuous) - GPS coordinates from onboard hardware - Historical occupancy logs and route records - Commuter natural-language queries (mobile app) - Live traffic and weather API feeds	Edge: YOLOv8-nano (Offline) detects passengers; Telemetry Packager handles online, lightweight cellular transmission of state data to the cloud.. Cloud Server: Gradient boosting ETA model; LSTM demand forecasting; route deviation and driving anomaly detection. NLP Chatbot: LLM API fusing occupancy + ETA for boarding recommendations	- LED strip color on windshield (Green / Yellow / Red / Blinking Red) - ETA and occupancy on commuter mobile app - Fleet dispatching alerts on operator dashboard - Safety anomaly alerts (operators first, users after verification)

2.2 Architecture Diagram:



SECTION 3: AI APPROACH & MODEL SELECTION

Field	Details
Primary AI Approach	[✓] Machine Learning [✓] NLP [✓] Computer Vision
Model Selection	• YOLOv8-nano (Ultralytics/PyTorch) offline edge passenger detection; deployed on Raspberry Pi 5 / Jetson Nano. • Facebook Prophet server-side time-series demand forecasting from historical occupancy logs. • Gradient Boosting Regressor (XGBoost) ETA prediction from GPS + traffic + weather features. • LLM API Lightweight cloud LLM — NLP boarding recommendation chatbot. • Stack: Python, C++, PyTorch, VS Code, FastAPI, React Native..
Reasoning	YOLOv8-nano is the only YOLO variant deployable within the thermal envelope of an unattended edge device, achieving real-time inference without cloud dependency. Gradient boosting generalizes better than deep ETA models during cold-start phases. Facebook Prophet was explicitly selected over LSTM for demand forecasting because it handles missing data and irregular reporting intervals gracefully—crucial during early pilot phases with sparse telemetry. The LLM chatbot is API-based to prevent edge compute overhead.

SECTION 4: DATA STRATEGY & ETHICS

Field	Details
Data Sources	• Custom collection: Overhead camera footage from pilot vehicles for YOLOv8 fine-tuning (collected under written operator consent) • GPS telemetry: Real-time position data from installed hardware • Open-source: OpenStreetMap (ODbL) for route geometry; MMDA/LTO historical traffic records for ETA model training • Weather: OpenWeatherMap API for ETA feature inputs • System-generated: Occupancy logs accumulated by LoadSense edge devices during pilot, bootstrapping the demand forecasting model
Data Quality & Cleaning	• GPS outliers removed via speed and heading sanity filters • Camera frames with severe occlusion or insufficient lighting flagged and excluded from training sets • Manual spot-check validation of occupancy counts during pilot phase • ETA predictions annotated with explicit confidence bounds during cold-start period
Licensing & Legality	• OpenStreetMap: ODbL permits use in this context. • OpenWeatherMap: Standard API TOS permits use. • In-vehicle camera footage: Collected under signed data collection agreements with vehicle operators. Footage is processed on-device; raw video is never transmitted or stored externally. Retained strictly for the pilot duration and deleted upon completion. • Custom GPS data: governed by written operator policy; no third-party sharing.

Bias Mitigation & Fairness	• Training data collected across multiple routes, time-of-day windows, and vehicle types to prevent route-specific or temporal bias • Occupancy classification is strictly threshold-based with no passenger profiling by appearance, demographics, or identity • Commuter recommendations are auditable by the transport authority to detect systematic disadvantage to specific operators or routes • Driver behavior data is advisory only; due-process review required before any disciplinary use — no automatic sanctions
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SECTION 5: DEVELOPMENT MILESTONES (AGILE ROADMAP)

Phase	Activity / Task	Tools Used	Expected Outcome
Sprint 1 (May 1-15)	• Camera test footage capture on pilot vehicle • Dataset labeling for passenger detection • YOLOv8-nano environment setup and initial inference • GPS hardware integration and telemetry endpoint	Python, GitHub, Roboflow, Raspberry Pi / Jetson Nano, FastAPI	Labeled dataset; verified hardware, telemetry pipeline
Sprint 2 (May 16-31)	• Fine-tune YOLOv8-nano on custom passenger dataset • Implement bidirectional line-crossing counter • GPIO LED strip integration for occupancy state output • GPS payload broadcast to backend server	VS Code, Ultralytics YOLOv8, PyTorch, FastAPI	Working edge AI occupancy counter; live GPS, occupancy broadcast
Sprint 3 (June 1-15)	• ETA model training (gradient boosting on GPS + traffic + weather) • Commuter mobile app UI with stop-level estimates • Operator dashboard with fleet dispatching view • AI chatbot boarding recommendation interface	React Native, Figma, scikit-learn / XGBoost, FastAPI, LLM AP	Working prototype: commuter app, operator dashboard, and chatbot
Sprint 4 (June 16-25)	• End-to-end integration testing • Edge case debugging (GPS dropout, low-light detection) • Demand forecasting model validation • Pitch preparation and demo video recording	GitKraken, Pytest, GitHub Actions	Final demo video; deployment-ready prototype

SECTION 6: SCALABILITY & REGIONAL RESILIENCE

Field	Details
Scaling Potential	LoadSense is hardware-agnostic, making ASEAN expansion a matter of procurement rather than architectural changes. Institutionally, initial scaling relies on partnering with local transport cooperatives under the LTFRB's PUV Modernization Program, positioning the system as a compliance-aligned add-on. Funding pathways include DOST-PCIEERD grants and smart city accelerator programs. The cloud layer scales horizontally, and cities with existing GTFS infrastructure can accelerate ETA training immediately. The commuter app targets low-RAM Android devices, the dominant device class across lower-income populations in Manila, Jakarta, and Ho Chi Minh City.
Technical Constraints	• HIGH — Operator Adoption: Requires hardware installation and driver acceptance. <i>Mitigation:</i> Engage through existing PUV cooperatives and introduce a revenue-sharing model on aggregated demand data to create clear economic incentives for operators. • HIGH — GPS Signal Degradation: Covered terminals cause dropouts. Dead-reckoning via accelerometer or cell-tower fallback is required . • MEDIUM — ETA Cold Start: Prediction accuracy requires at least two weeks of accumulated operational data; early estimates carry explicit uncertainty margins . • MEDIUM — Driver Hardware Tampering: Intentional hardware disconnection is flagged by the server as a signal anomaly distinct from standard GPS dropout; physical mitigations for confirmed tampering will be finalized during pilot deployment, as of now there is no full way to prevent/avoid tampering. • LOW — Mobile Rendering: Full interactive maps are infeasible on low-RAM Android. Tile-based static maps are used — a correct trade-off for the target user.

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INSTRUCTIONS FOR SUBMISSION:

1. **Strictly PDF:** Convert your final document to PDF.
2. **Visuals Matter:** Use clear diagrams for Section 2.
3. **Submission Link:** Team Leaders must upload this file to the **Official Submission Folder**

► **SUBMISSION - TECHNICAL ROADMAP** by **May 17, 2026** (9:00 PM GMT+7).